MATH-121 LA & ODEs

Week | Lecture 9

~~5/5/12~~ 20/5/21

Exact First Order ODEs

In general $\frac{dy}{dx} = F(x,y) \rightsquigarrow (1)$

If we can write (1) as

$$M(x,y)\,dx + N(x,y)\,dy = 0 \rightsquigarrow (2)$$

and
$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Then the differential equation is exact

Method of solution

Step I:- (1) Integrate function $M(x,y)$ w.r.t 'x'

(2) Integrate only those terms of function $N(x,y)$ which are free from 'x' with respect to y.

(3) Add above two expressions and put them equal to constant.

Example

$$\frac{dy}{dx} = -\frac{(2xy+y^2+3)}{x^2+2xy} \rightsquigarrow (1)$$

$$\boxed{(2xy+y^2+3)\,dx + (x^2+2xy)\,dy = 0}$$

$M(x,y) = 2xy+y^2+3$ $\qquad$ $N(x,y) = x^2+2xy$

$\frac{\partial M}{\partial y} = 2x+2y$ $\qquad$ $\frac{\partial N}{\partial x} = 2x+2y$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Therefore, the differential eq is exact.

Solution

(1) $\rightarrow \int (2xy+y^2+3)\,dx$

(2) $\rightarrow \underset{\text{free from 'x'}}{\int} (0)\,dy$

(3) ~~$\int (2xy$~~

(3) $\rightarrow$ $\int (2xy+y^2+3)\,dx + \int 0\,dy = C_0$

$$\frac{\cancel{2}x^2y}{\cancel{2}} + y^2x + 3x + C_1 = C$$

$D = C - C_1$

$$\boxed{x^2y + y^2x + 3x = D}$$

Ans

MATH-121 LA & ODEs 12/5/2020

Review Exact First Order ODEs

In general $\frac{dy}{dx} = f(x,y) \rightsquigarrow (1)$

If we can write ODEs as

$$M(x,y)\,dx + N(x,y)\,dy = 0 \rightsquigarrow (2)$$

and $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

then the differential equation (1) or (2) is exact.

Example $(3x^2y + 2)dx + (x^3 + y)dy = 0 \rightsquigarrow (1)$

$M(x,y) = 3x^2y + 2$ $N(x,y) = x^3 + y$

$\frac{\partial M}{\partial y} = 3x^2 \rightsquigarrow$ $\frac{\partial N}{\partial x} = 3x^2$

Eq (1) is exact ODEs.

Solution:-

(1) $\rightarrow \int M(x,y)\,dx = \int (3x^2y + 2)\,dx$

(2) $\rightarrow \underset{\text{free of } x}{\int N(x,y)\,dy} = \int y\,dy$

(3) $\rightarrow \int (3x^2y + 2)\,dx + \int y\,dy = C$

$$= \frac{\cancel{3}x^3}{\cancel{3}}y + 2x + \frac{y^2}{2} = C$$

$$\boxed{2x^3y + 4x + y^2 = D} \quad \text{Ans.} \qquad D = 2C$$

Exercise

i) $\dfrac{(3y\cos 3x\,dx - \sin 3x\,dy)}{y^2} = 0$

ii) $x\,dx + y\,dy + \dfrac{x\,dy + y\,dx}{x^2+y^2} = 0$

iii) $(r\cos^2\theta - \sin\theta)\,dr - r\cos\theta(r\sin\theta + 1)\,d\theta = 0$

iv) $\left[\dfrac{3-y}{x^2}\right]dx + \left[\dfrac{y^2-2x}{xy^2}\right]dy = 0 \;;\; y(-1) = 2$

iv) $x\,e^{x^2-y^2}\,dx = y\left[e^{x^2-y^2} + 1\right]dy \;;\; y(0) = 0$

Linear ODEs of First Order

The general form

$$\boxed{\frac{dy}{dx} + P(x)\,y = Q(x)}$$

Example:-

$$\frac{dy}{dx} = 2xy \rightsquigarrow (1)$$

$$\Rightarrow \quad \frac{dy}{dx} - 2xy = 0 \rightsquigarrow (2)$$

In eq (2) $\quad P(x) = -2x$ and $Q(x) = 0$

Example (2)

$$xy' + y = x + x^3$$

$$y' + \frac{1}{x}y = 1 + x^2$$

$$P(x) = \frac{1}{x} \; ; \; Q(x) = 1 + x^2$$

Example (3)

$$y' + (\tanh x)y = \sinh 2x$$

$$P(x) = \tanh x \quad Q(x) = \sinh 2x.$$

Method of solution

Given $\boxed{\frac{dy}{dx} + P(x)\, y = Q(x)}$ ⟶ (1)

Step I :- Find Integrating factor

$$I.F = e^{\int P(x)\,dx} \quad \rightarrow (2)$$

Multiply eq (1) with Integrating factor in (2)

This will reduce (1) into ~~seperable~~ form differential

— — — — — —

Example (1)

$$(x-1)^3 \frac{dy}{dx} + 4(x-1)^2 y = x+1$$

$$\frac{dy}{dx} + \frac{4\cancel{(x-1)^2}}{\cancel{(x-1)^3}}\, y = \frac{x+1}{(x-1)^3}$$

$$\frac{dy}{dx} + \left[\frac{4}{x-1}\right] y = \frac{x+1}{(x-1)^3} \quad \rightarrow (1)$$

$$P(x) = \frac{4}{x-1} \qquad Q(x) = \frac{x+1}{(x-1)^3}$$

5

$$I.F. = e^{\int P(x)\,dx} = e^{\int \frac{4}{x-1}dx} = e^{4\int \frac{1}{x-1}dx}$$

$$= e^{4\ln(x-1)} = \cancel{e}^{\cancel{\ln}(x-1)^4} = (x-1)^4 \rightsquigarrow 2$$

Multiply 1 with 2

$$(x-1)^4\frac{dy}{dx} + \frac{4(x-1)^{\cancel{4}}}{\cancel{x-1}}y = \frac{x+1}{\cancel{(x-1)^3}}(x-1)^{\cancel{4}}$$

$$(x-1)^4\frac{dy}{dx} + 4(x-1)^3 y = (x+1)(x-1)$$

$$\boxed{\frac{d}{dx}\left[(x-1)^4 y\right] = x^2-1}$$

Integrate both sides w.r.t x

$$\int \frac{\cancel{d}}{\cancel{dx}}\left[(x-1)^4 y\right]\cancel{dx} = \int x^2-1\,dx$$

$$(x-1)^4 y = \frac{x^3}{3} - x + C$$

$$y = \frac{1}{(x-1)^4}\left[\frac{x^3}{3} - x + C\right]$$

Ans.

~~Ex 2~~

$$y' - \frac{2}{x}y = x^2e^x \rightsquigarrow 1$$

$$P(x) = \frac{-2}{x} \qquad Q(x) = x^2e^x$$

$$I.F. = e^{\int P(x)dx} = e^{\int \frac{-2}{x}dx} = e^{-2\int \frac{1}{x}dx} = e^{-2\ln x} = \cancel{e}^{\cancel{\ln} x^{-2}}$$

$$I.F. = x^{-2} \rightsquigarrow 2$$

Multiply 1 with 2

$$\frac{1}{x^2}\frac{dy}{dx} - \frac{2}{x}\left(\frac{1}{x^2}\right)y = \frac{\cancel{x^2}e^x}{\cancel{x^2}}$$

$$\frac{1}{x^2}\frac{dy}{dx} - \frac{2}{x^3}y = e^x$$

$$\boxed{\frac{d}{dx}\left[\frac{1}{x^2}y\right] = e^x}$$

Integrate w.r.t 'x'

$$\int \frac{\cancel{d}}{\cancel{dx}}\left[\frac{1}{x^2}y\right]\cancel{dx} = \int e^x$$

$$\frac{1}{x^2}y = e^x + C$$

$$\boxed{y = x^2e^x + Cx^2}$$ Ans.

$y' \sin x + y \cos x = \cos x \; ; \quad y(\pi/2) = 0$

$$y' + \frac{\cos x}{\sin x} y = \frac{\cos x}{\sin x}$$

$$\boxed{y' + \cot x \, y = \cot x} \rightsquigarrow 1$$

$P(x) = \cot x \qquad Q(x) = \cot x$

$$I.F. = e^{\int P(x)dx} = e^{\int \cot x \, dx} = e^{\int \frac{\cos x}{\sin x} dx} = \cancel{e}^{\cancel{\ln} \sin x}$$

$$= \sin x \rightsquigarrow 2$$

Multiply 1 with 2

$$\sin x \frac{dy}{dx} + \cot x \sin x \, y = \cot x \sin x$$

$$\frac{dy}{dx} \sin x + \frac{\cos x}{\cancel{\sin x}} \cancel{\sin x} \, y = \frac{\cos x}{\cancel{\sin x}} \cancel{\sin x}$$

$$\sin x \frac{dy}{dx} + \cos x \, y = \cos x$$

$$\boxed{\frac{d}{dx}[\sin x \, y] = \cos x}$$

Integrate w.r.t 'x'

$$\cancel{\int} \frac{\cancel{d}}{dx}[\sin x \, y] \, d\cancel{x} = \int \cos x \, dx$$

$$\sin x \, y = \sin x + C$$

$$y = \frac{\cancel{\sin x}}{\cancel{\sin x}} - \frac{c}{\sin x}$$

$$\boxed{y = 1 - c\,\mathrm{cosec}\,x} \rightsquigarrow (4)$$

Now using $y(\pi/2) = 0$

that is $\boxed{x = \pi/2 \,;\, y = 0}$ in (4)

$$0 = 1 - c\,\mathrm{cosec}(\pi/2) \qquad \mathrm{cosec}(\pi/2) = 1$$

$$0 = 1 - c$$

$$\Rightarrow \boxed{c = 1}$$

Use in (4)

$$\boxed{y = 1 - \mathrm{cosec}\,x}$$

Ans.